Customer Segmentation

961731 Business Data Analytics

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Module 1: Strategic Motivation

Connecting cross-domain market inefficiencies to enterprise capital allocation

Market Dynamics and WTP

- Purchasing Power
  - Capital distribution varies across consumer segments.
- Willingness to Pay
  - Consumers express different price sensitivities for products.
- Value Capture
  - Segmented pricing captures consumer surplus and increases margin.

Premium Price Segment (High WTP)
Margin Maximization Area

Mid-Tier Segment (Average WTP)
Volume/Margin Balance Area

Economy Segment (Low WTP)
Volume Driver Area

Systemic Inefficiencies: Wasted Marketing Spend

- Customer Heterogeneity
  - Generic products fail to satisfy distinct user preferences.
- Wasted spend:
  - Traditional mass marketing targets undifferentiated audiences, leading to inefficient capital allocation.
- Segment fragmentation:
  - Customers exhibit distinct buying preferences that require targeted outreach channels.

Mass Market
(Undifferentiated)

Single Product Profile
(Low conversion / High waste)

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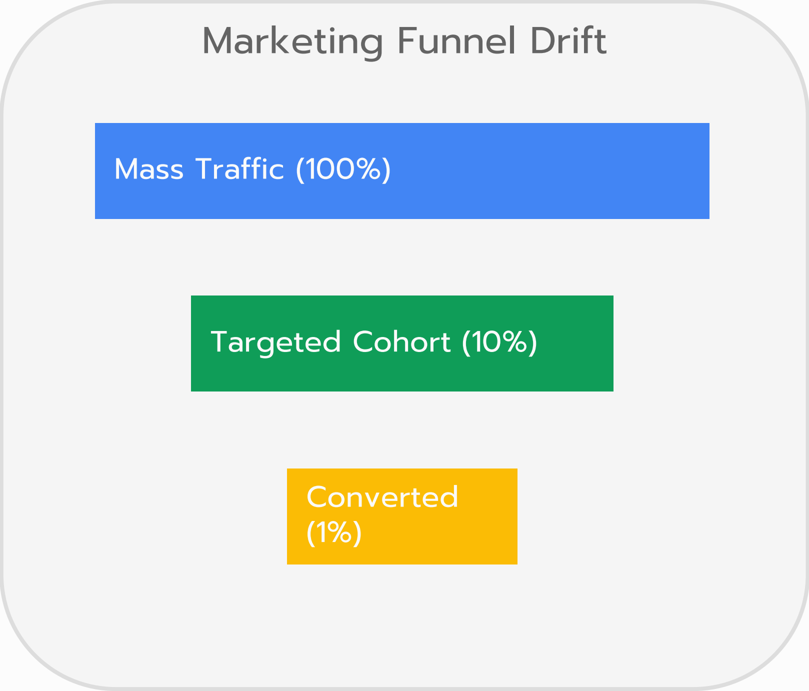
Systemic Inefficiencies: Diluted Customer Lifetime Value

- Generic campaigns:
  - Single-message campaigns dilute customer interest and lead to high churn rates.
- Value erosion:
  - Failing to identify high-value cohorts reduces long-term margin potential.



The Market Inefficiency Problem: Customer Heterogeneity

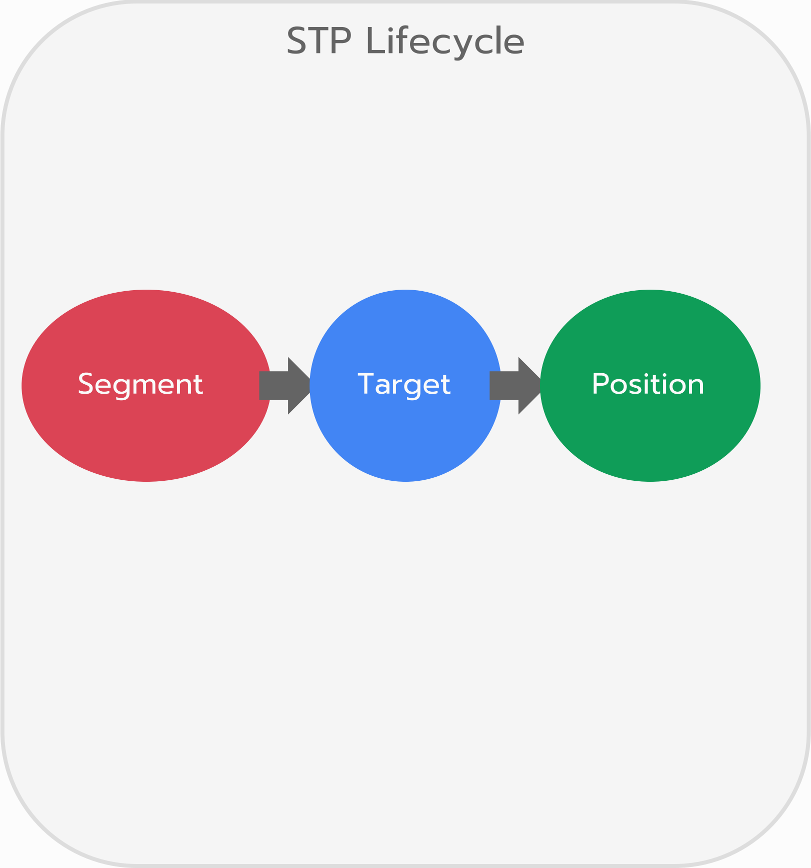
- Diverse preferences:
  - Customers differ in price sensitivities, brand loyalty, and product interaction patterns.
- Resource leakage:
  - Over-serving low-value cohorts leaks operational capacity from high-margin segments.

Concept Definition: Market Segmentation

- Market Segmentation:
  - Dividing a heterogeneous market into distinct, homogeneous customer subgroups.
- Target Marketing:
  - Selecting one or more customer segments to target with specific offerings.
- Product Positioning:
  - Establishing a distinct brand image in the customer's mind.

Concept Definition: STP Framework

- Strategic alignment:
  - Aligning segmentation directly with product design and pricing strategies.
- Competitive advantage:
  - Positioning products uniquely against competitors to capture maximum market share.

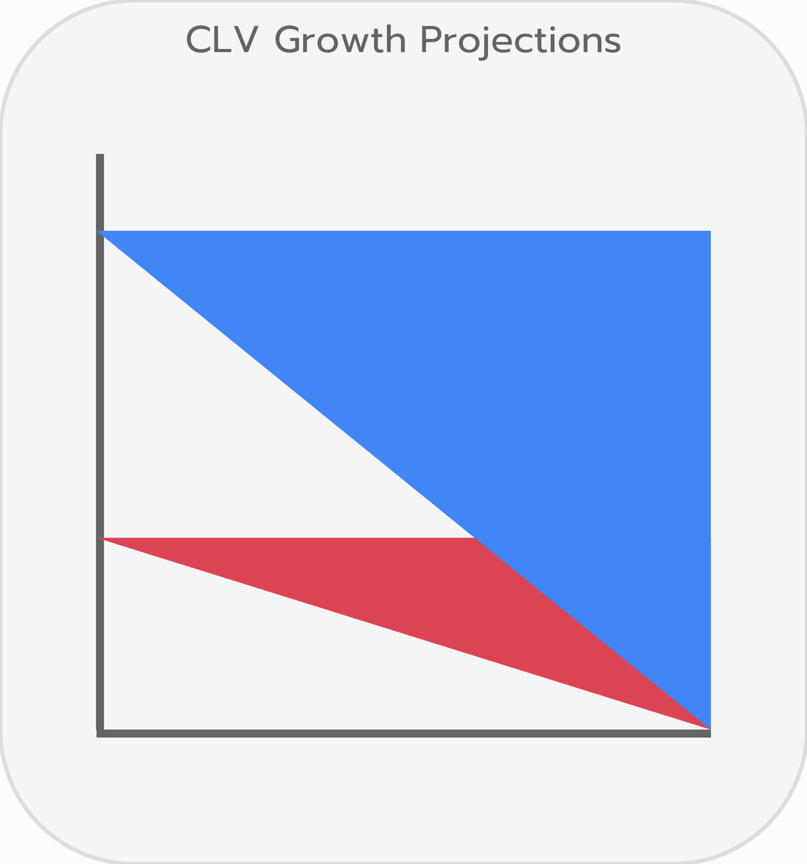


Module 2: Value Vectors

Quantifying segmentation impact on customer acquisition and capital efficiency

Value Vector: CLV Optimization

- Retention programs:
  - Custom loyalty programs for high-value segments increase retention and cumulative margin.
- Expansion revenue:
  - Cross-selling and upselling tailored offers increases the average order value.

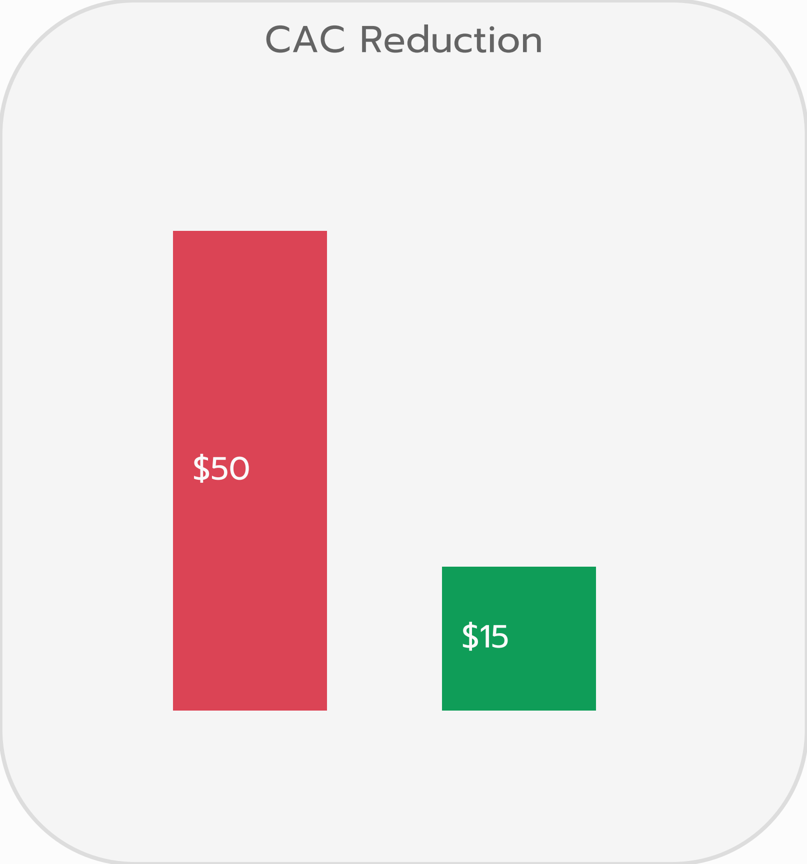


Market Segmentation Variables

- Geographic Variables
  - Partition market by country, state, city, or neighborhood density.
- Demographic Variables
  - Segment based on age, gender, income, race, and education.
- Psychographic Variables
  - Analyze social class, lifestyle, and customer personality traits.
- Behavioral Variables
  - Group by occasion, loyalty, user rate, and attitude.

Value Vector: Acquisition Cost Reduction

- Target channels:
  - Allocating marketing resources to high-conversion channels reduces overall acquisition costs.
- Campaign precision:
  - Personalized messaging increases ad response and click-through rates.



Value Vector: Product Positioning & Tailoring

- Custom value:
    - Creating product bundles customized for distinct segment preferences.
- Market share:
    - Addressing underserved niches captures margins that competitors overlook.

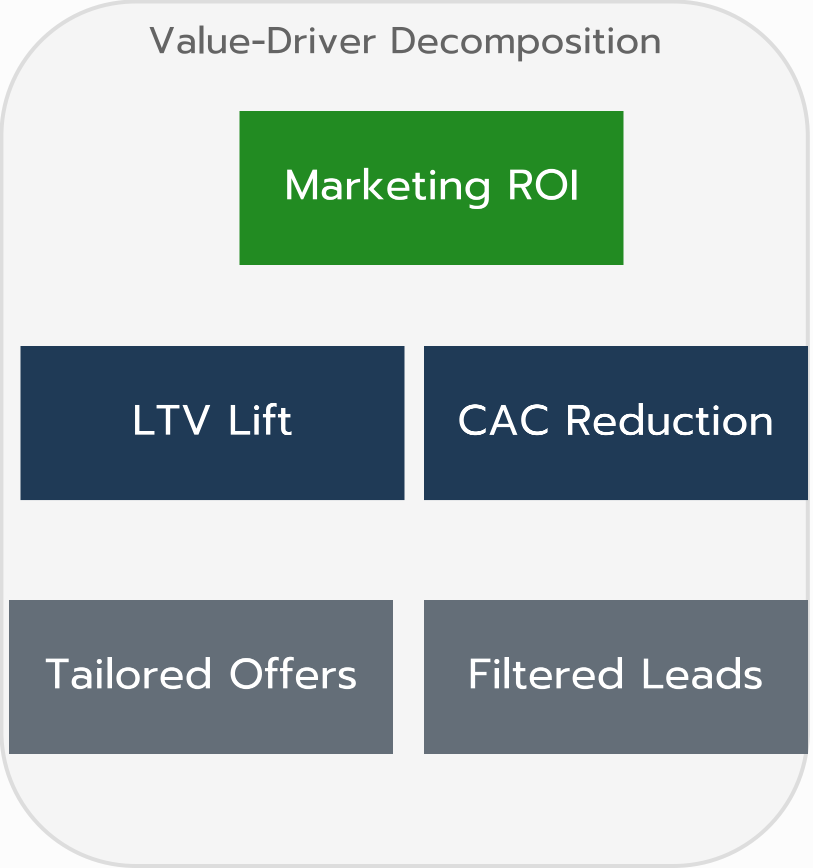
Segment	Key Feature
Budget Conscious	Discount Bundles
Premium Loyalists	VIP Service

Value Vector: Dynamic Pricing & Margin Capture

- Elasticity variance:
  - Different customer segments have varying price elasticities.
- Surplus capture:
  - Charging premium prices to low-elasticity segments captures consumer surplus.

Resource Allocation Optimization

- CAC Efficiency
  - Concentrating capital on receptive segments minimizes customer acquisition cost.
- LTV Maximization
  - Nurturing high-value segments increases long-term profit contribution.
- Resource Allocation
  - Unsegmented spending yields a **capital depletion risk**.



Module 3: Traditional Methodologies

Exploring heuristic-based customer grouping and its structural limitations

Baseline Dataset

- Sample data:
  - An 8-customer baseline dataset used to contrast traditional heuristics with clustering.
- Dimensions:
  - Features include Age, Annual Income, and Spending Score.

CustID	Age	Income	Spend
C001	19	15k	39
C002	21	15k	81
C003	20	16k	6
C004	23	16k	77
C005	31	95k	15
C006	32	97k	79
C007	35	98k	17
C008	34	100k	82

Traditional Demographic Sorting

- Demographic Sorting
  - Traditional methods sort customers using manual, single-dimension rules.
- Gender Splits
  - Customers are separated into Female and Male segments.
- Age Cutoffs
  - Arbitrary age divisions (e.g. Under 35 vs 35+) partition groups.

Females
(Age Under 35: 1)
(Age 35+: 4)

Males
(Age Under 35: 1)
(Age 35+: 1)

Traditional Method: Demographic Binning

- Binning heuristics:
  - Grouping customers by age brackets or arbitrary income intervals.
- Hidden patterns:
  - Binning hides high-spending young customers among low-spending peers.

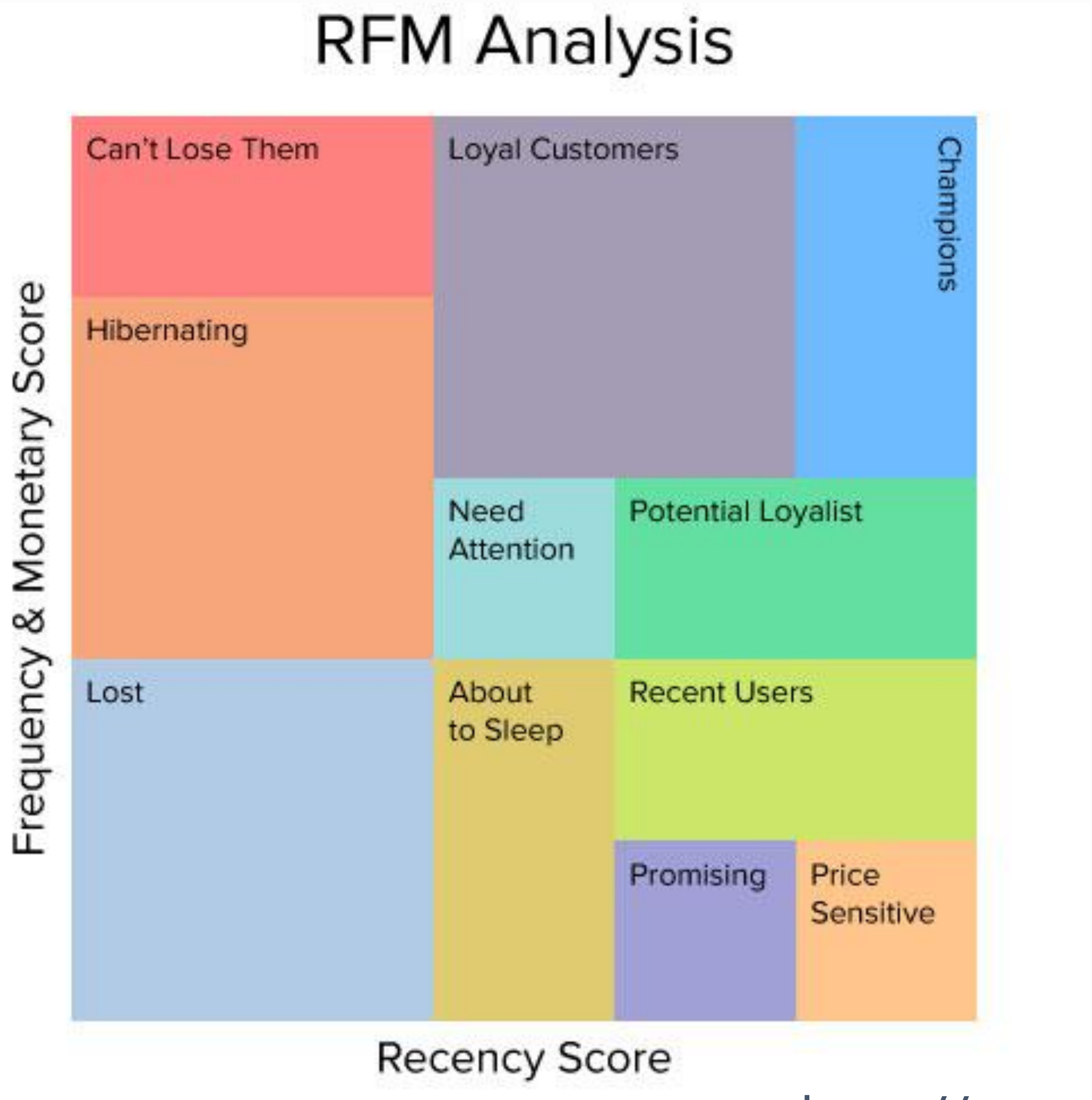
Age Bin	Income Bin	Segment
18-25	Low	Impulsive Youth
26-50	High	Frugal Professional
51+	Low	Budget Senior

Traditional Method: RFM Scoring Matrix

- Manual scoring:
  - Recency, Frequency, and Monetary scores are binned manually.
- Fixed boundaries:
  - Arbitrary thresholds fail to adapt to shifts in customer behavior.

Recency	Frequency	Monetary
High (5)	High (5)	Champions
Low (1)	High (5)	Loyalists
High (5)	Low (1)	At Risk

Traditional Method: RFM Scoring Matrix



https://www.nvecta.com/blog/rfm-analysis/

Combinatorial Explosion

- Feature Growth
  - Adding more features exponentially increases the number of sub-segments.
- Branch Complexity
  - With D features and C categories, total segments = C^D.
- Subsegment Dilution
  - High feature counts dilute customer counts, leading to an **empty segment risk**.

Variables (D=5) & Categories (C=4)
Total Segments = 4^5 = 1,024 Groups
Most groups will contain zero customers!

Statistical Limitations

- Arbitrary Thresholds
  - Manual cutoffs (e.g. Income > $50k) ignore continuous variation.
- Boundary Errors
  - A customer earning $49k is grouped differently than one earning $50k.
- Information Loss
  - Discretizing continuous features discards valuable variance data.

Income Boundary Cutoff
($50,000)
Customer A: $49,999 -> Economy
Customer B: $50,001 -> Premium
High similarity, forced separation!

Structural Limitation

- Linear Assumptions
  - Traditional sorting assumes linear relationships between features.
- Interaction Neglect
  - Non-linear feature interactions (e.g., Age impacting Spending) are missed.
- Strategy Blindspots
  - Neglecting interactions leads to a **campaign misalignment risk**.

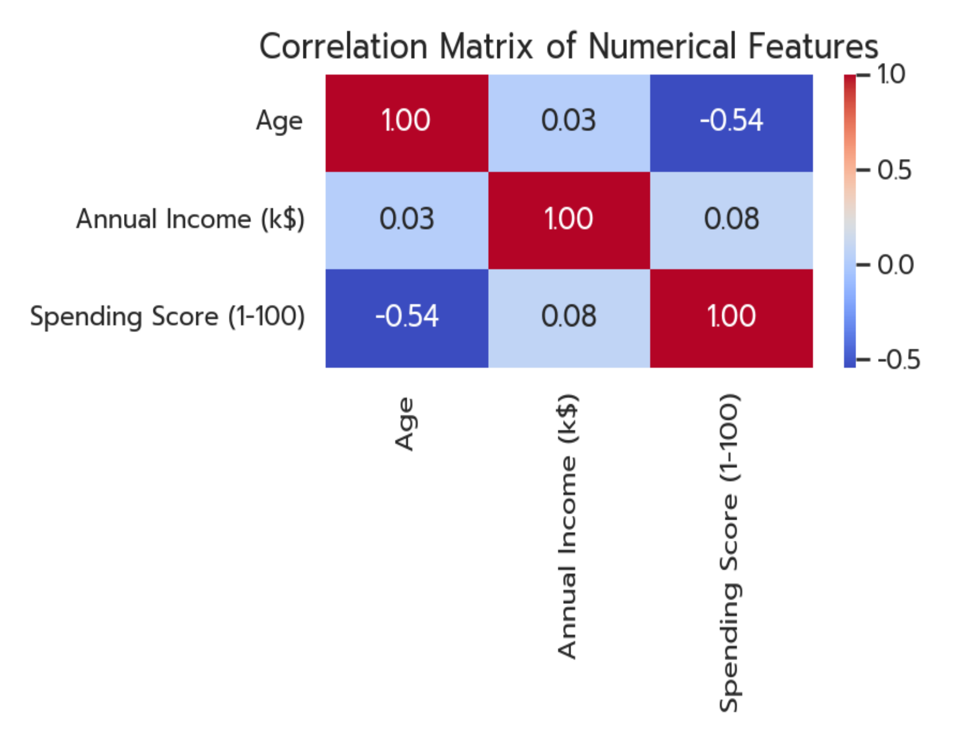
Interactive Feature Space
Age and Income interact non-linearly.
Simple cutoffs fail to isolate the true high-value clusters.

Module 4: Machine Learning Paradigm

Transitioning from heuristics to unsupervised clustering algorithms

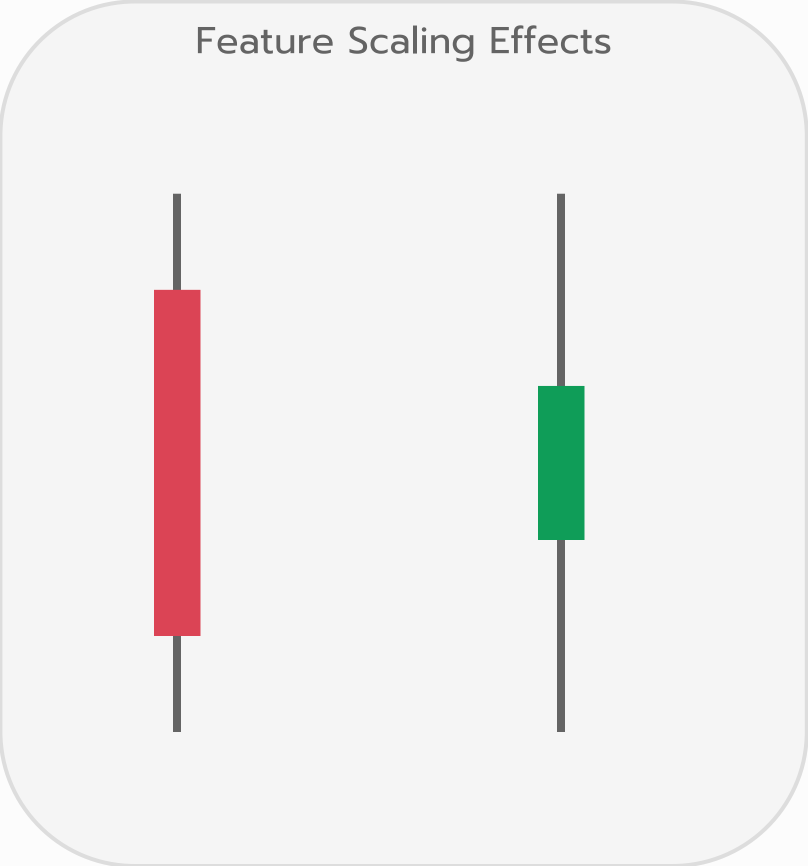
Transition to Clustering

- Unsupervised Learning
  - Letting algorithms identify clusters based on actual feature similarity.
- Correlation Matrix
  - EDA heatmap shows relationships between Age, Income, and Spending.
- Interpretation
  - Weak correlation suggests multidimensional clustering is required.



Exploratory Data Analysis (EDA) for Clustering

- Feature scaling:
  - Features with larger scales dominate distance calculations and require normalization.
- Normal distribution:
  - Visualizing distributions identifies outliers that could skew centroids.



Importance of Standardization

- Distance Sensitivity
  - K-Means uses Euclidean distance, which is sensitive to feature scales.
- Feature Dominance
  - Unscaled Income ($15k-$137k) dominates over Age (18-70) in calculation.
- Z-Score Scaling
  - Scaling features to mean=0 and variance=1 prevents scale dominance.

Standardization Formula
x_scaled = (x - mean) / std
Ensures all dimensions contribute equally to distance metrics.

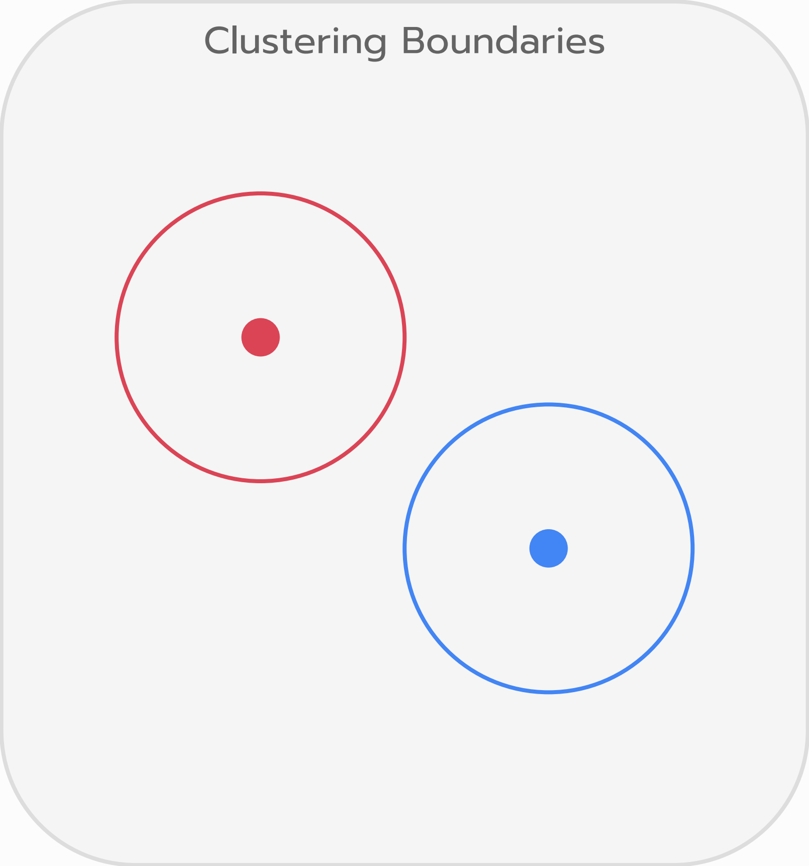
Machine Learning Concept: Unsupervised K-Means

Data-driven borders:

Letting the algorithm find natural boundaries rather than imposing arbitrary rules.

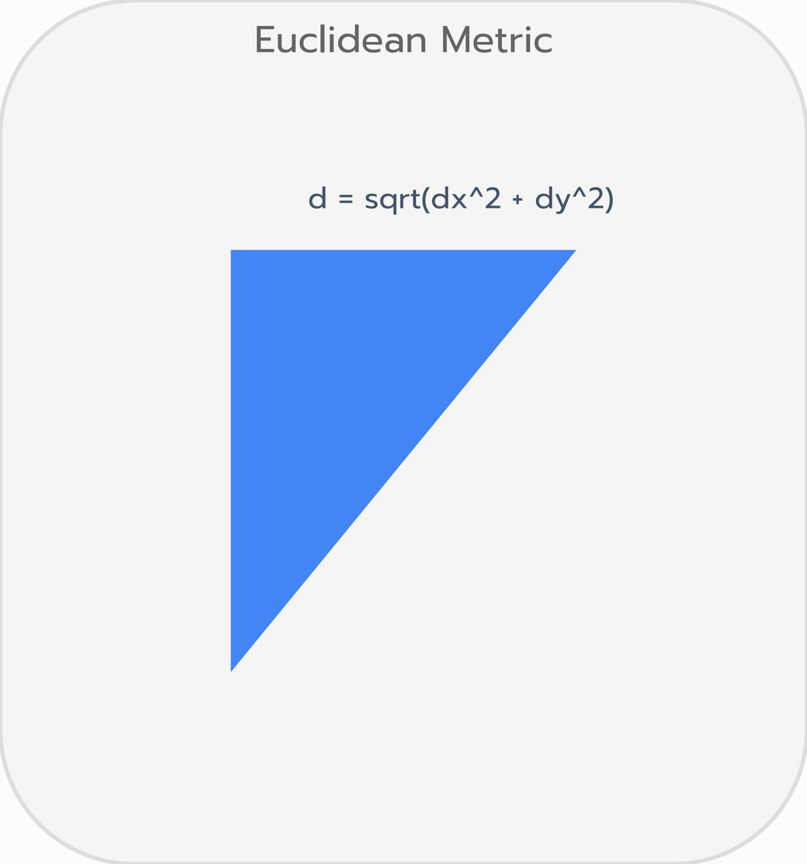
Centroid positioning:

- Clusters are defined by central representative points in multi-dimensional space.



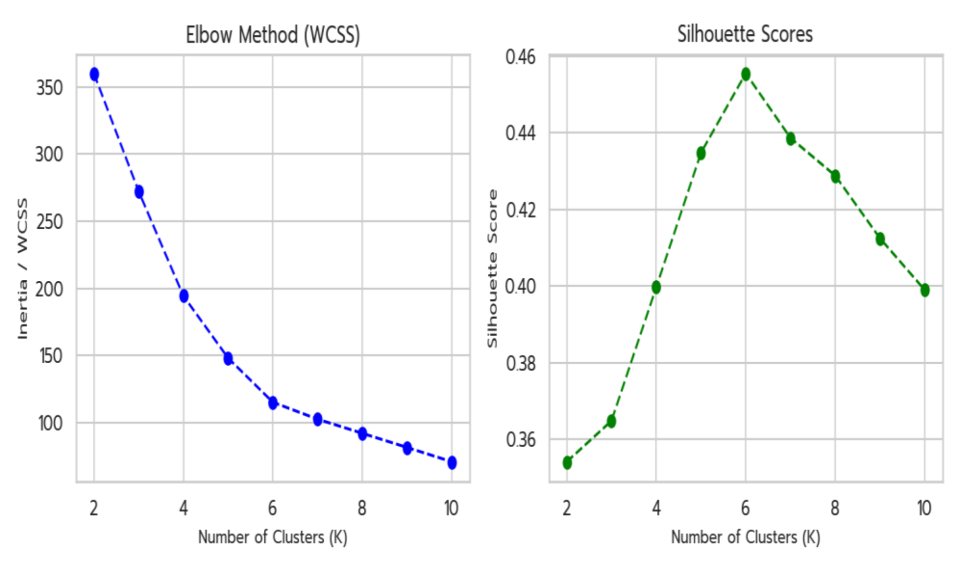
Metric 1: Euclidean Distance

- Distance metric:
  - Measuring geometric distance between customer points in N-dimensional space.
- Formula:
  - Square root of the sum of squared differences across features.
- Business meaning:
  - Shorter distances represent higher behavioral similarity between customers.



Metric 2: Within-Cluster Sum of Squares (WCSS)

- Cluster compactness:
    - Sum of squared distances from all points in a cluster to their centroid.
- Elbow method:
    - Plotting Within-Cluster Sum of Squares (WCSS) against cluster count reveals the point of diminishing returns.
- Business meaning:
    - Lower WCSS indicates tight, highly-focused customer groupings.



Metric 3: Silhouette Coefficient

- Cluster separation:
  - Measuring how close a point is to its own cluster compared to others.
- Score range:
  - Ranges from -1 to +1, where values near +1 indicate excellent separation.
- Business meaning:
  - High scores confirm distinct customer personas with minimal profile overlap.

Rule Interpretation & Strategic Deployment

- Profiling centroids:
  - Analyzing average feature values of each cluster to define customer personas.
- Campaign mapping:
  - Directing specific campaigns to clusters based on their profiled needs.

Centroid ID	Income Profile	Target Campaign
Cluster 0	High Income	Luxury Invite
Cluster 1	Low Income	Budget Coupons

Rule Interpretation & Deployment

- Cluster 0: Frugal
  - High Income, Low Spending. Action: Offer high-end premium bundles.
- Cluster 1: Sensible
  - Low Income, Low Spending. Action: Retain via essentials and value pricing.
- Cluster 2: VIP
  - High Income, High Spending. Action: Enroll in Elite loyalty concierge program.
- Cluster 3: Spendthrift
  - Low Income, High Spending. Action: Deploy micro-promos and monitor credit risk.

Strategic Segment Allocation
Frugal: Cross-sell high-end products
VIP: Exclusive loyalty program
Average: Mid-tier nurture campaign

Module 5: Pedagogical Board Demonstration

Step-wise mathematical walkthrough of K-Means clustering

Board Demo: Step 1 - Centroid Initialization

- Seed selection:
  - Choosing initial centroid coordinates to start the clustering process.
- Selected seeds:
  - Set Centroid 1 at Point A and Centroid 2 at Point C.

Step 1: Init Centroids

Centroid 1 (C1)
Customer 2: (15, 2, 40)

Centroid 2 (C2)
Customer 5: (3, 5, 220)

Board Demo: Step 2 - Distance Calculation & Assignment

- Distance check:
  - Calculating Euclidean distance from all points to both centroids.
- Assignment rule:
  - Assigning points to the nearest centroid. A and B go to C1.

Point	Distance to C1	Distance to C2	Assignment
Cust 1	111.9	71.2	Cluster 2
Cust 2	0.0	180.4	Cluster 1
Cust 3	92.4	91.2	Cluster 2

Board Demo: Step 3 - Centroid Recalculation

- Mean calculation:
  - Updating centroids as the average coordinates of all assigned points.
- Updated points:
  - Centroid 1 shifts to (15.5, 79.0). Centroid 2 shifts to (96.0, 47.0).

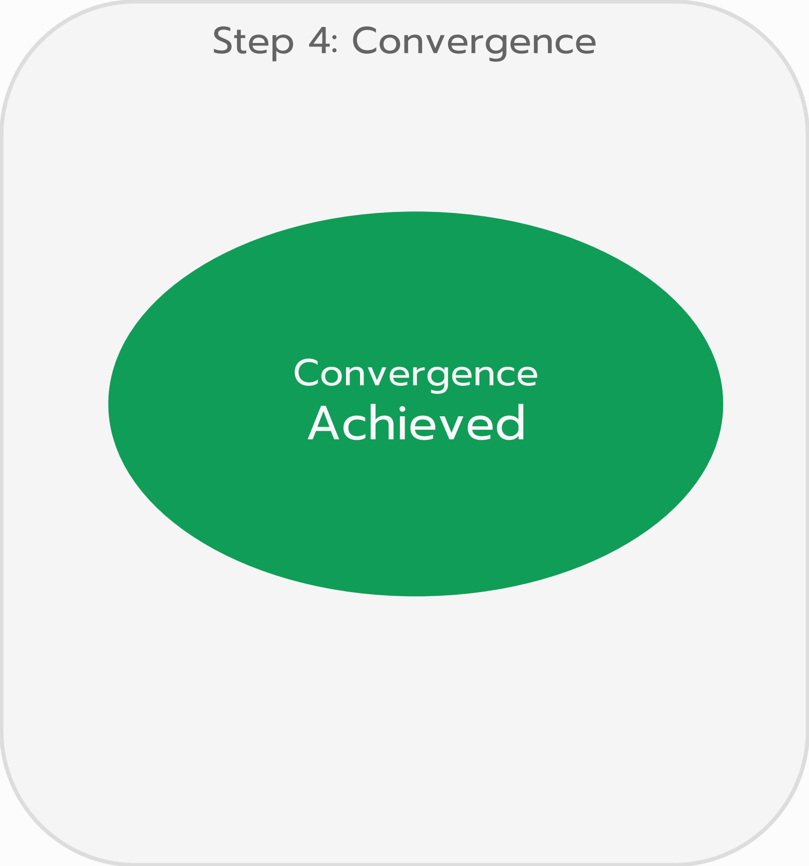
Step 3: Update Centroids

New Centroid 1 (C1)
Mean coordinates: (23.3, 2.0, 40.0)

New Centroid 2 (C2)
Mean coordinates: (4.0, 5.5, 210.0)

Board Demo: Step 4 - Iteration & Convergence

- Stability check:
  - Reassigning points based on new centroids and checking for changes.
- Convergence:
  - Cluster assignments remain identical. Centroids are stable and the algorithm stops.



Final Cluster Profiles

- Cluster 1 profile:
  - Low Income, High Spend customer cohort.
- Cluster 2 profile:
  - High Income, Low Spend customer cohort.

Profile ID	Size	Monetary Centroid
C1 (Frugal)	3 members	40.00
C2 (VIP)	2 members	210.00
C3 (Loyal)	3 members	146.67

Module 6: Programmatic Implementation

Translating mathematical logic into enterprise-grade Python code

Data Preprocessing Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler

# Ingest and scale data
df = pd.read_csv('mall_customers.csv')
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[[
    'Age',
    'Annual Income (k$)',
    'Spending Score (1-100)'
]])
```

Lines 1-2:

Import pandas and StandardScaler modules.

Line 5:

Ingest retail database CSV into DataFrame.

Lines 6-7:

Standardize variables to remove scale dominance.

K-Means Clustering Code

```
from sklearn.cluster import KMeans

# Fit K-Means with K=5
kmeans = KMeans(
    n_clusters=5,
    random_state=42,
    n_init='auto'
)
df['Cluster'] = kmeans.fit_predict(X_scaled)
print(df.groupby('Cluster').mean())
```

Line 1:

Import KMeans module from scikit-learn.

Lines 4-8:

Instantiate KMeans clustering with 5 clusters.

Line 9:

Fit model and predict cluster assignments.

Cluster Visualization Code

```
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

# Reduce to 2D with PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
plt.scatter(
    X_pca[:, 0], X_pca[:, 1],
    c=df['Cluster'], cmap='Set1'
)
plt.show()
```

Lines 1-2:
: Import plotting and PCA packages.

Lines 5-6:
: Project 3D features into 2D PCA space.

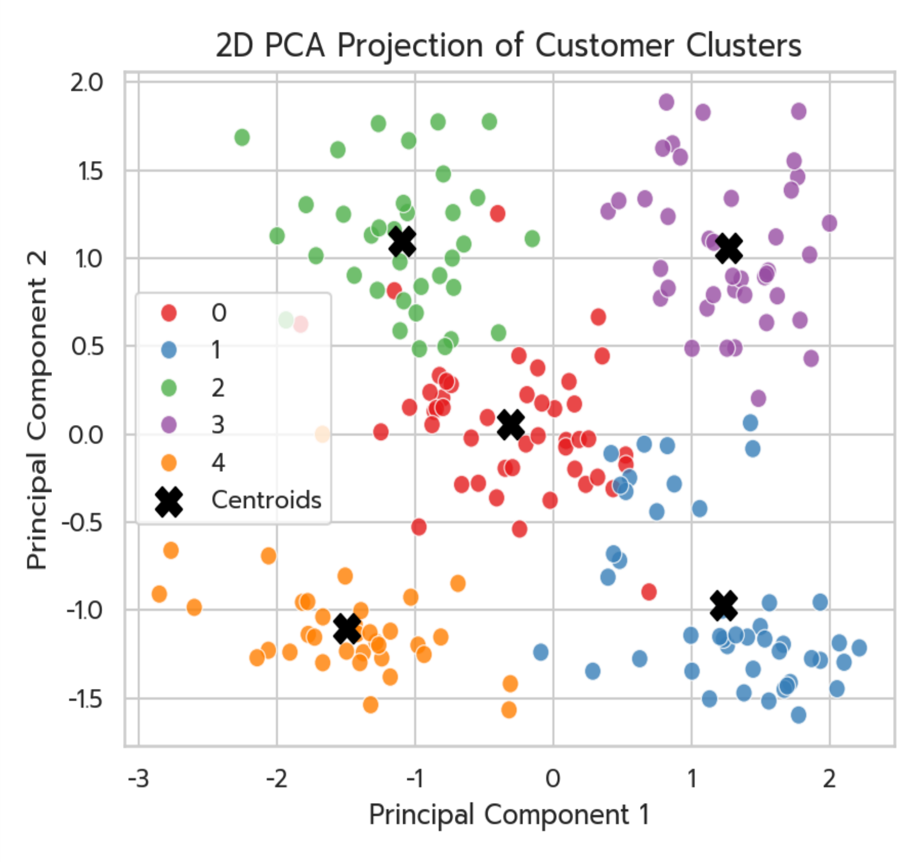
Lines 7-10:
: Generate scatter plot color-coded by cluster.

Process Visualization Pipeline

- Pipeline Flowchart
  - Steps to convert raw transaction records into strategic customer segments.

- Standardization Input
  - Scale features to normalize distance contribution.

- Visualization Outcome
  - Analyze cluster projection to define segment-specific promotions.



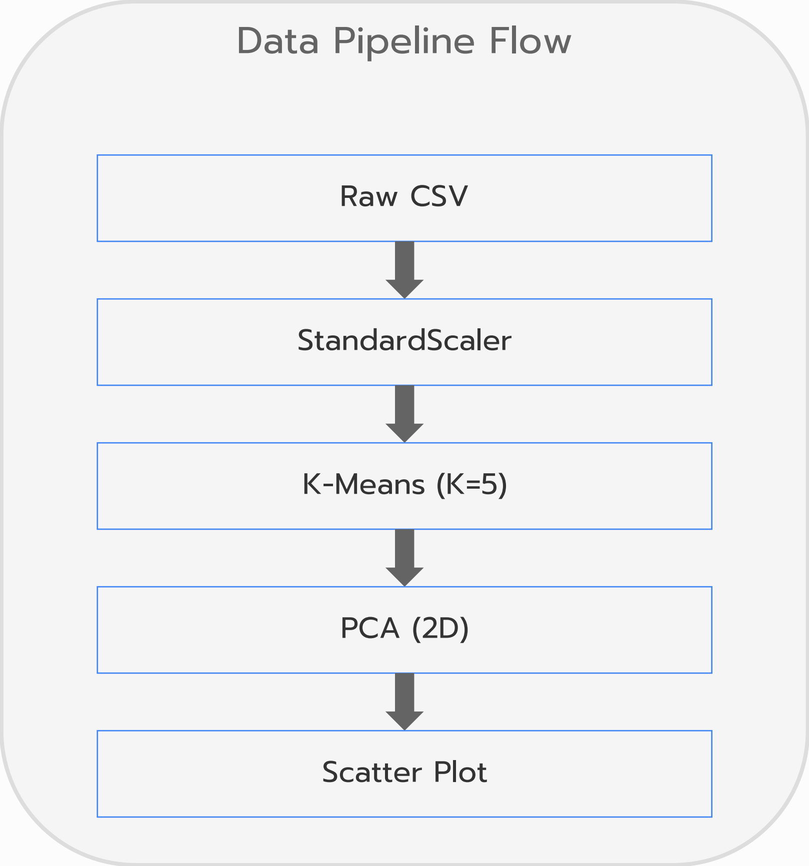
Process Visualization

Data flow:

Processing steps from raw CSV to scaled PCA visualization.

Model validation:

Automated checkpoints verify model convergence and silhouette metrics.



Module 7: Strategic Interpretation

Converting model outputs into actionable business decisions

Strategic Decisions: Type 1 vs Type 2

Type 1 decisions:
- Irreversible physical changes such as shelf planogram layout redesign.

Type 2 decisions:
- Reversible digital promotions like email coupons and dynamic site banners.

Parameter	Type 1 Decision	Type 2 Decision
Reversibility	Irreversible	Highly Reversible
Cost	High Capital	Low Operational
Scope	Physical Store	Digital Campaign

Type 1: VIP Retention Program

- VIP Churn Mitigation
  - Pareto ratio 80:20.
- Elite Loyalty Program
  - Launch exclusive loyalty concierge service to mitigate competitor attrition.
- Financial Risk
  - Failing to retain VIP segment presents a **significant revenue loss risk**.

Identify VIP Segment (K=5 Model)

Auto-Enroll in Concierge Service

Reduce Churn rate

Type 2: Virtual Product Bundles

- Digital Personalization
  - Virtual bundling offers are easily modified in response to conversion.
- VIP Premium Bundles
  - Bundle luxury products with high-margin accessories at checkout.
- Frugal Value Bundles
  - Offer multi-buy discounts to frugal customers to increase basket size.

Virtual Checkout Screen Mockup
[Item 1: Premium Cologne]
[Recommended Bundle: Silk Tie - 20% Off]
[Add Bundle to Cart - One Click]

Type 2: Targeted Digital Campaigns

- Digital Campaigns
  - Reversible target marketing campaigns tailored to low-spending groups.
- Spendthrift Offers
  - Send high-frequency, low-value coupons to younger spendthrifts.
- Default Risk
  - Over-promoting credit to low-income segments poses a **bad debt default risk**.

Digital Ad Trigger: Low-Spend Group

Distribute 20% Off Mobile Coupon

Measure Conversion & Default Rates

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Strategic Governance: Managing by Fact

- Model drift:
  - Customer behaviors shift over time, requiring monthly cluster reassessment.
- Performance tracking:
  - Tracking campaign margin lift against baseline non-targeted groups.

A/B Test Dashboard

Variant A (Targeted Campaigns)
Gross Margin Lift: +18.4%

Variant B (Control Group)
Gross Margin Lift: +0.2%

Module 8: Case Studies & Business Simulations

Applying customer segmentation to real-world datasets

Case Study 1: Manual Cafe Transaction Log

- Construct RFM and cluster using Python

- The transactions are generated based on four underlying behavioral customer archetypes:
  - VIP Daily Drinkers: High frequency, low recency (frequent, recent visits), and higher spend per transaction (buying multiple items/meals).
  - Budget Students: High/medium frequency, low spend per visit (buying cheap drip coffee or muffins), low recency.
  - Occasional Treaters: Low/medium frequency, high spend per visit, average recency.
  - Churned Customers: Low frequency and very high recency (only visited in the first half of the year).
- Data file : Cafe_Transactions_example.csv
  - Unique Customers: 150 loyalty members (CUST-001 through CUST-150)
  - Columns: Customer, Total Spend, Visits Date time (formatted as YYYY-MM-DD HH:MM:SS), and Item (realistic coffee shop menu items with base prices)
  - Null Values: 0 nulls across all columns

Case Study 2: Programmatic Mall Segmentation

- (Very) Large-scale dataset
- Size: Million rows (I promised you !!!!)
- Key points:
  - Scale: Optimize code, work with chunking, or use big data frameworks like PySpark or Dask.
  - Behavioral RFM: Calculate advanced features, such as the recency of click views versus actual purchase events, cart abandonment rates, and total spending.

Dataset Access URL

https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store